INDEPENDENCE OF THE GYSIN-AT-INFINITY CLASS FROM THE CHOICE OF THE PROPER MODEL, UP TO THE INDUCED MORPHISM AT INFINITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FAQ (sources/stacks-chow/pinned-original-source.tex line 10108).

Allowed standard prerequisites (editorial).

- The Gysin-at-infinity bivariant class and its defining property. Same chapter, Lemma labelled lemma-gysin-at-infty (source line 10026), invoked in the statement. Supplies the two classes C and C' and the characterisation C cap alpha = i_infinity^* beta used to compare them.
- Compatibility of the Gysin homomorphism with proper pushforward. Same chapter, Lemma labelled lemma-closed-in-X-gysin (source line 5264), invoked at proof line 10125. Supplies i_infinity^* beta = g_{infinity,*}(i'_infinity)^* beta', which is the identity that makes the two constructions agree.

Source reference. Original lemma 0F9J: sources/stacks-chow/pinned-original-source.tex, source lines 10026–10026. Same-chapter pointer into the author environment lemma at source line 10026; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02TA: sources/stacks-chow/pinned-original-source.tex, source lines 5264–5264. Same-chapter pointer into the author environment lemma at source line 5264; invoked at proof line 10125 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Lemma 0F9J let $g : W' \to W$ be a proper morphism which is an isomorphism over $\mathbf{A}^1_X$. Let $C' \in A^0(W'_\infty \to X)$ and $C \in A^0(W_\infty \to X)$ be the classes constructed in Lemma 0F9J. Then $g_{\infty,*} \circ C' = C$ in $A^0(W_\infty \to X)$.*

Proof. Set $b' = b \circ g : W' \to \mathbf{P}^1_X$. Denote $i'_\infty : W'_\infty \to W'$ the inclusion morphism. Denote $g_\infty : W'_\infty \to W_\infty$ the restriction of g. Given $\alpha \in \mathrm{CH}_k(X)$ choose $\beta' \in \mathrm{CH}_{k+1}(W')$ restricting to the flat pullback of α on $(b')^{-1}\mathbf{A}^1_X$. Then $\beta = g_*\beta' \in$

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

$\mathrm{CH}_{k+1}(W)$ restricts to the flat pullback of α on $b^{-1}\mathbf{A}^1_X$. Then $i_\infty^*\beta = g_{\infty,*}(i'_\infty)^*\beta'$ by Lemma 02TA. This and the corresponding fact after base change by morphisms $X' \to X$ locally of finite type, corresponds to the assertion made in the lemma. $\square$